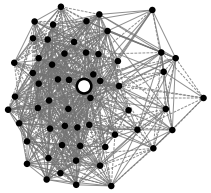
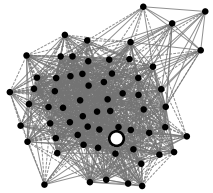
DATA PREPARATION AND PATIENT SPLIT

ABIDE I, 871 patient scans

control patient



ASD patient



SLIDING-WINDOW BRAIN GRAPH BUILDING

30-TR windows, stride 10; Pearson correlation; signed graph, top 20% edges

BRAIN-REGION SUBGRAPH TRACKING

20 selected brain regions (ROIs), 1-hop neighbourhood; node set fixed at the first window

GIN ENCODING AND GRU NEXT-WINDOW PREDICTION

2-layer weighted GIN, mean pool, $z \in \mathbb{R}^{128}$; GRU (hidden 128), MSE + variance loss, controls only

TEMPORAL SUBGRAPH ANOMALY SCORING FOR ASD DETECTION

patient score = max over regions and transitions

CONTROL-BASED DETECTION THRESHOLD CALIBRATION

95th percentile of validation-control scores (transition and patient level)

SYNTHETIC SUBGRAPH ANOMALY GENERATION

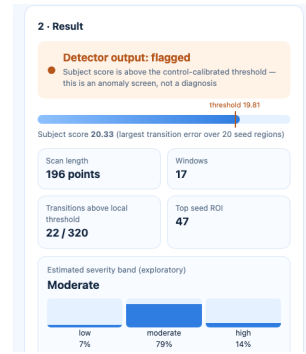
Gradient-guided PGD edge-weight edits, 5 steps, up to 15% of edge pairs, 3 severities

FEW-SHOT ASD ADAPTATION

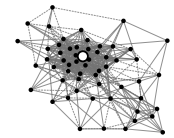
1 / 5 / 10 ASD patient scans + 30 generated anomalies; logistic regression on 20-D region scores

ANOMALY DETECTION AND SEVERITY OUTPUT

Flag if patient score exceeds threshold τ; ADOS-2 band (low / moderate / high) by logistic regression



real control



synthetic anomaly

